CSA5108 Cryptography and Network security

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Experiment 3 : Playfair cipher

Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char matrix[5][5];

void createMatrix(char *key) {
   int used[26] = {0};
   int row = 0, col = 0;

   for (int i = 0; key[i] != '\0'; i++) {
      char c = toupper(key[i]);

      if (c == 'J')
         c = 'I';

      if (c >= 'A' && c <= 'Z' && !used[c - 'A']) {
         matrix[row][col++] = c;
         used[c - 'A'] = 1;

         if (col == 5) {
            col = 0;
            row++;
         }
      }
   }

   for (char c = 'A'; c <= 'Z'; c++) {
      if (c == 'J')
```

```
            continue;

        if (!used[c - 'A']) {
            matrix[row][col++] = c;
            used[c - 'A'] = 1;

            if (col == 5) {
                col = 0;
                row++;
            }
        }
    }
}

void findPosition(char c, int *row, int *col) {
    if (c == 'J')
        c = 'I';

    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++) {
            if (matrix[i][j] == c) {
                *row = i;
                *col = j;
                return;
            }
        }
    }
}

void prepareText(char *input, char *output) {
    int k = 0;

    for (int i = 0; input[i] != '\0'; i++) {
        if (isalpha(input[i])) {
            char c = toupper(input[i]);

            if (c == 'J')
                c = 'I';
```

```
            output[k++] = c;
        }
    }

    output[k] = '\0';

    char temp[1000];
    int n = 0;

    for (int i = 0; i < k;) {
        char a = output[i];
        char b;

        if (i + 1 < k)
            b = output[i + 1];
        else
            b = 'X';

        temp[n++] = a;

        if (a == b) {
            temp[n++] = 'X';
            i++;
        } else {
            temp[n++] = b;
            i += 2;
        }
    }

    if (n % 2 != 0)
        temp[n++] = 'X';

    temp[n] = '\0';
    strcpy(output, temp);
}

void encrypt(char *text) {
```

```
    for (int i = 0; text[i] != '\0'; i += 2) {
        int r1, c1, r2, c2;

        findPosition(text[i], &r1, &c1);
        findPosition(text[i + 1], &r2, &c2);

        if (r1 == r2) {
            text[i] = matrix[r1][(c1 + 1) % 5];
            text[i + 1] = matrix[r2][(c2 + 1) % 5];
        } else if (c1 == c2) {
            text[i] = matrix[(r1 + 1) % 5][c1];
            text[i + 1] = matrix[(r2 + 1) % 5][c2];
        } else {
            text[i] = matrix[r1][c2];
            text[i + 1] = matrix[r2][c1];
        }
    }
}

int main() {
    char key[100];
    char plaintext[1000];

    printf("Enter keyword: ");
    fgets(key, sizeof(key), stdin);
    key[strcspn(key, "\n")] = '\0';

    printf("Enter plaintext: ");
    fgets(plaintext, sizeof(plaintext), stdin);

    createMatrix(key);

    printf("\nPlayfair Matrix:\n");
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++)
            printf("%c ", matrix[i][j]);
        printf("\n");
    }
```

```
    prepareText(plaintext, plaintext);
    encrypt(plaintext);

    printf("Ciphertext: %s\n", plaintext);

    return 0;
}
```

